

Question number	Answer	Additional guidance	Mark
5(a)(i)	113		(1)

Question number	Answer	Additional guidance	Mark
5(a)(ii)	An explanation that includes the following points: (albumin / R groups) {negatively charged / polar / hydrophilic} (1) - and will therefore dissolve in the {water / blood <u>plasma</u>} (1) - forms hydrogen bonds with water (1)	ACCEPT water / plasma is a polar solvent ACCEPT description of H bonds / weak bonds / Van der Waals forces DO NOT ACCEPT ionic bonds	(2)

Question number	Answer	Additional guidance	Mark
5(b)(i)	1.175 (g dm⁻³)	ACCEPT 3 / 3.2 / 3.24 (times higher)	(1)

Question number	Answer	Additional guidance	Mark
5(b)(ii)	An explanation that includes the following points: • (thickest band) albumin is the most concentrated (in this sample) (1) • (moved the furthest) albumin could be the most soluble (1) • in the solvent used (1) • (moved the furthest) albumin could be the {lowest mass / lightest} (1)	ACCEPT more albumin present ACCEPT most (negatively) charged	(3)

Question number	Answer	Additional guidance	Mark
5(c)	An explanation that includes the following points: • 15 $\mu\text{mol dm}^{-3}$. is too low to activate the channel / 500 $\mu\text{mol dm}^{-3}$ will activate the channel (1) • therefore sperm will not be activated in the {seminal fluid / male} / therefore sperm activated in the {uterus / female} (1) • this will avoid waste of {resources / named resource} (1) • so more {viable / healthy} sperm (1) • so more likely fertilisation will take place (1)	ACCEPT refs to high or low concentrations ACCEPT fewer sperm activated / more sperm activated prevent early activation sperm can swim faster / acrosome enzymes will be released / sperm ready to digest the egg cell membrane e.g. energy / ATP / acrosome ACCEPT successful	(4)

